Augmented Human Lab: Inspiring curiosity for science in kids with Google Cloud solutions

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About Augmented

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About Augmented Human Lab

Augmented Human Lab at Auckland Bioengineering Institute develops human-machine interfaces to enhance life and help people with disabilities. The lab spearheads Kiwrious project, a project that deploys digital technology to inspire children with a passion and curiosity for science. Kiwrious won the social impact category in the University of Auckland's $100K Velocity Challenge for social and environmental entrepreneurship.

Google Cloud results

- Enables creative endeavor by saving development team eight hours per week on operational tasks
- Facilitates growth of education program by accelerating ethics approvals through world-class data security
- Enables open-ended development of education platform with versatile range of IoT and ML solutions

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Industries: Education

Location: New Zealand

Cloud Functions

Google Cloud (https://cloud.google.com/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Load Balancing (https://cloud.google.com/load-balancing/)

Cloud Functions (https://cloud.google.com/functions/)

(https://cloud.google.com/container-registry/),

Cloud SQL (https://cloud.google.com/sql/)

AI Platform (https://cloud.google.com/ai-platform/)

Cloud Build (https://cloud.google.com/build/)

Cloud IoT Core (https://cloud.google.com/iot-core/)

Cloud Storage (https://cloud.google.com/storage/)

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Container Registry (https://cloud.google.com/container-registry/)

Chromebook (https://chromeenterprise.google/devices/)

Google Workspace (https://workspace.google.com/)

- **Promotes education mission by enabling 250-fold increase in platform queries by students**

Suranga Nanayakkara founded the Augmented Human Lab (https://www.ahlab.org/) on a vision to create life-enhancing, human-machine interfaces, such a wearable device (https://www.ahlab.org/project/aisee) that enables the blind to access information about objects by pointing to them. The research center that's part of the Auckland Bioengineering Institute (https://www.auckland.ac.nz/en/abi.html) (at the University of Auckland (https://www.auckland.ac.nz/en.html)) fosters technology as "a natural extension of our body, mind, and behavior."

When Suranga came across a global study that showed 45% of New Zealand's 8th graders lack confidence in science subjects, he sensed an ideal opportunity to make a difference through his vision of human-friendly machine intelligence. Could his team create a fun computer interface that becomes an extension of the child's curiosity and imagination?

"Our mission is to change the way children learn science around the world. In Google Cloud we met a kindred spirit. Tools like Cloud Build, Cloud Load Balancing, and Google Kubernetes Engine enable us to deploy fast and scale instantly to keep pace with the growth of pupils' minds."

—Suranga Nanayakkara, Associate Professor, Augmented Human Lab, Auckland Bioengineering Institute

The Kiwrious (https://kiwrious.com/) project was born. It makes science playful and exciting by extending the power of smart devices with plug-in sensors that capture environmental phenomena, including temperature, ultraviolet (UV) light, humidity, and air quality. Turning scientific inquiry into an adventure, it encourages kids to make their own discoveries rather than grapple exclusively with formulas and charts.

To spread the Kiwrious mission of empowerment, especially in disadvantaged schools, Suranga needed powerful cloud solutions to free up his small team from mundane tasks such as server maintenance, enable unlimited scaling power, and protect the identity of pupils with world-class cloud security. Google Cloud (https://cloud.google.com/) delivered a multiplier effect for the Kiwrious equation to inspire children with a passion for science.

"Our mission is to change the way children learn science around the world. In Google Cloud we met a kindred spirit," says Suranga. "Tools like Cloud Build (/build), Cloud Load Balancing (/load-balancing), and Google Kubernetes Engine (/kubernetes-engine) (GKE) enable us to deploy fast and scale instantly to keep pace with the growth of pupils' minds."

Easy-to-use solutions for auto-deployment and unlimited scaling

To draw children into the Kiwrious concept, Suranga's team invented a character from outer space (called Zally) who has difficulty tolerating UV rays. Pupils use Google Chromebooks (/chrome-enterprise/chromebooks) and Kiwrious plug-in sensors to create environments that minimize UV, with methods such as sunscreen and plastic enclosures.

What happened next surprised the researchers. Pupils began spontaneously inventing their own experiments, such as finding out what flower scents bees are attracted to (using a volatile organic compound sensor).

The simplicity of Kiwrious sensors on Chromebooks, says Suranga, is key to encouraging the spirit of inquiry. For the Kiwrious team, Google Cloud's intuitive interface and user-friendly documentation have been instrumental to making highly complex coding tasks, involving machine learning (ML) and IoT, feel like a fun adventure.

"As soon as we make a request to trigger new code through Cloud Build, we can see it being deployed, tested, and instantly reaching users. Google Cloud abstracts all of these tasks away from us. That means we can focus on building the platform rather than maintaining cloud solutions."

—Alaeddin Nassani, Research Fellow, Augmented Human Lab, Auckland Bioengineering Institute

Auckland Bioengineering Institute had already been using Google Workspace (https://workspace.google.com/) for internal communications. So the transition to Google Cloud was a natural one when the team decided to move to a cloud environment from an on-premises virtual machine setup, seeking greater scalability and faster code deployment.

Appreciating its powerful simplicity, Google Cloud became an obvious option for the team. "Google Cloud comes out of the box with sample code that we can easily click on to build. Our team uses that functionality to prototype quickly and get code up and running," says Alaeddin Nassani, an Augmented Human Lab research fellow who drives IT operations.

The Kiwrious development team codes on GitHub and uses Cloud Build for automated deployment to its Terraform (infrastructure-as-code) architecture. GKE then combines with Cloud Load Balancing's intelligent global resource distribution to enable seamless and affordable scaling, both for the frontend student experience and backend server-side code.

Cloud Build's CI/CD (continuous integration/continuous deployment) solution, which automates deployment of code, has been important in freeing up the small Kiwrious team of 18 to focus on creative tasks such as building software, says Alaeddin.

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Concrete results that bring confidence in exponential growth

The operational gains from adopting Google Cloud solutions have been significant. When Kiwrious was on an on-premises architecture, it took the team two days to deploy a single software release. Now in a cloud environment built around Google Cloud, the team pushes out code within 10 minutes. Deploying Cloud Build alone saves the DevOps team an estimated eight hours per week in monitoring and maintenance tasks.

Kiwrious is still in beta stage and has reached about 200 students in a handful of schools. That's about to change virtually overnight. Through a grant from the New Zealand Ministry of Business Innovation and Employment, Augmented Human Lab is preparing to deploy Kiwrious into a nationwide pilot program reaching roughly 5,000 students in early 2021.

Student queries to the platform are expected to soar from about 400 to roughly 100,000 per day, about a 250% increase from its current load. If successful, the program will be rolled out internationally with an anticipated exponential growth in demand.

Suranga says that with Google Cloud, he's confident the team is ready. "Google Cloud's scaling, code deployment, and load balancing tools mean we'll be able to handle whatever load surges that come our way," he says. To further enhance Kiwrious capabilities, he envisions a transition to a completely serverless environment, using Cloud Functions (/functions) as the core solution.

Securing sensitive ethics approvals through Google Workspace

Google Workspace has long driven the Auckland Bioengineering Institute's collaborative ethos, enabling secure file-sharing and real-time teamwork between hubs in New Zealand, Singapore, and Sri Lanka, where Suranga is from. Now it's meeting a new critical need for the Kiwrious project: securing student data.

Because the program deals with children's personal information, one challenge is to obtain ethics approvals to expand the project's scope. Google Workspace's robust security has been instrumental in resolving the issue. By using secure Google Forms (https://workspace.google.com/products/forms/) for student questionnaires, Kiwrious is able to put the minds of parents, teachers, and administrators at ease about the security of personal data.

"At the university, we need to get the ethics approvals," says Suranga. "When we told administrators the personal information was going to be collected in a secure setup using Forms and

Google Drive (https://workspace.google.com/products/drive/), with all of the rock-solid security protocols, it smoothed the way to get the approvals we needed to get the project off the ground."

"The possibilities are endless. Since our mission at Augmented Human Lab is to enrich the human-machine relationship, we're excited to be developing solutions using Google Cloud that nourish young minds and feed curiosity through technology with a human touch."

—Suranga Nanayakkara, Associate Professor, Augmented Human Lab, Auckland Bioengineering Institute

Future of unlimited scientific inquiry with ML and IoT solutions

Suranga's team has had its eye on a limitless Kiwrious future from the start. Beyond New Zealand, it's soon planning to launch in Singapore, Sri Lanka (where it hopes to empower rural children), and eventually globally.

A key advantage of being on Google Cloud is the unlimited potential to evolve Kiwrious' intelligent capabilities with ML. A future step would be to adopt IoT Core (/iot-core) to connect multiple sensors in far-flung schools and deploy AI Platform (/ai-platform) to improve project design through predictive modeling of student behavior. Google Cloud IoT and ML tools will eventually enable Kiwrious to create global collaborative student experiments using data from Kiwrious sensors.

"The possibilities are endless," says Suranga. "Since our mission at Augmented Human Lab is to enrich the human-machine relationship, we're excited to be developing solutions using Google Cloud that nourish young minds and feed curiosity through technology with a human touch."